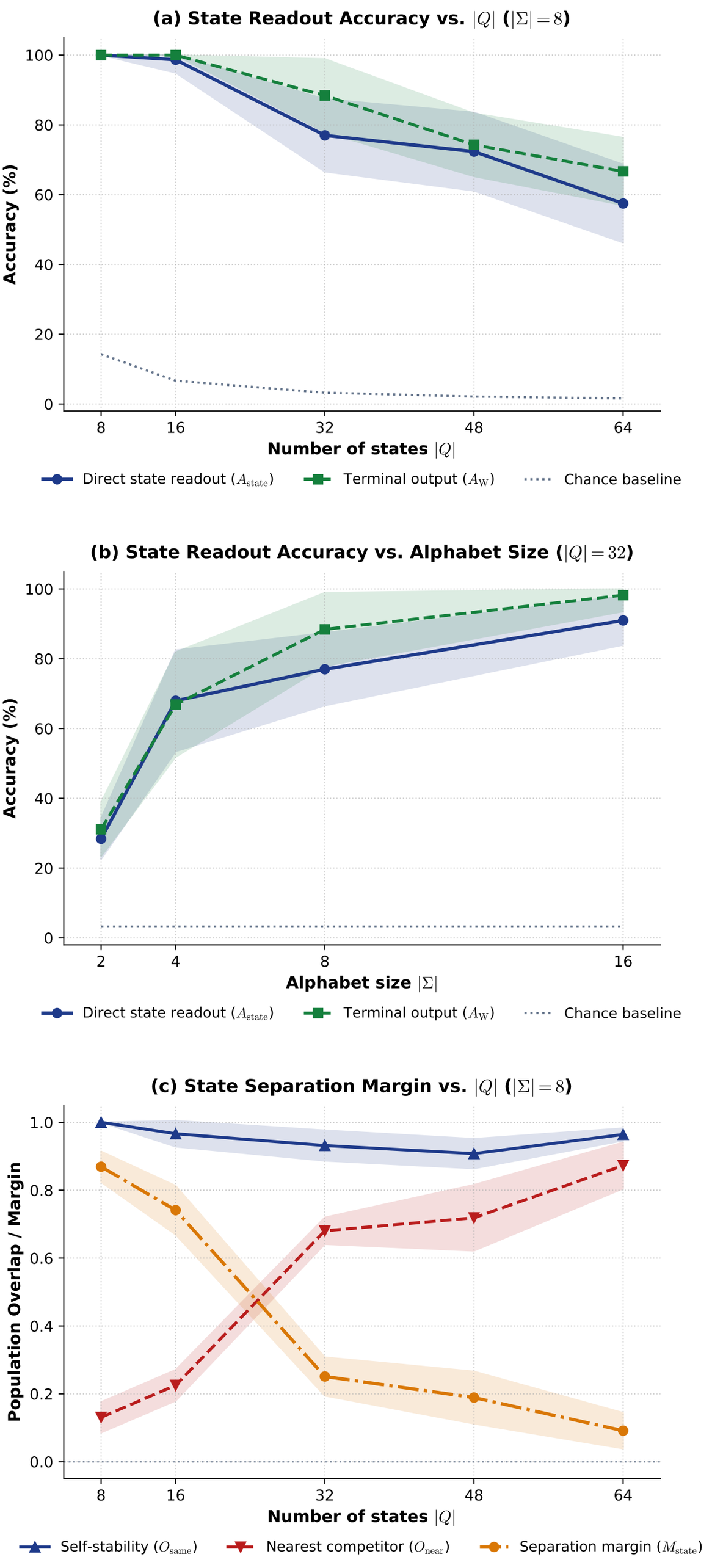
(a) State Readout Accuracy vs. $|Q|$ ($|\Sigma| = 8$)
Accuracy (%)
100
80
60
40
20
0
8
16
32
48
64
Number of states $|Q|$
Direct state readout (A_{state})
Terminal output (A_{W})
Chance baseline
(b) State Readout Accuracy vs. Alphabet Size ($|Q| = 32$)
Accuracy (%)
100
80
60
40
20
0
2
4
8
16
Alphabet size $|\Sigma|$
Direct state readout (A_{state})
Terminal output (A_{W})
Chance baseline
(c) State Separation Margin vs. $|Q|$ ($|\Sigma| = 8$)
Population Overlap / Margin
1.0
0.8
0.6
0.4
0.2
0.0
8
16
32
48
64
Number of states $|Q|$
Self-stability (O_{same})
Nearest competitor (O_{near})
Separation margin (M_{state})